

Java Program Skeleton (Beginner Template)

// Program Name: _____

// Author: _____

// Date: _____

```
import java.util.*;
```

```
public class ClassName
```

```
{
```

```
    // Global Variables (Optional)
```

```
    // User Defined Methods (Optional)
```

```
    public static void methodName()
```

```
    {
```

```
        // Method Body
```

```
    }
```

```
    // Main Method
```

```
    public static void main(String[] args)
```

```
    {
```

```
        // Variable Declarations
```

```
        // Input Statements
```

```
        // Processing Logic
```

```
        // Output Statements
```

```
    }
```

```
}
```

Simple Skeleton for Most Programs

```
public class ClassName
```

```
{
```

```
    public static void main(String[] args)
```

```
    {
```

```
        // Write your code here
```

```
        System.out.println("Hello World");
```

```
    }
```

```
}
```

Skeleton with Scanner Input

```
import java.util.Scanner;
```

```
public class ClassName
```

```
{
```

```
    public static void main(String[] args)
```

```
    {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        // Input
```

```
        int num = sc.nextInt();
```

```
        // Processing
```

```
        int square = num * num;
```

```
// Output
System.out.println("Square = " + square);

    sc.close();
}
}
```

Note: Replace ClassName with your actual class name. The file name must be the same as the public class name (e.g., ClassName.java).

Java Program Structure for Beginners

A Java program is generally written in the following structure:

// 1. Documentation / Comments

// 2. Package Declaration (Optional)

// package mypackage;

// 3. Import Statements (Optional)

import java.util.*;

// 4. Class Declaration

public class HelloWorld

{

// 5. Main Method

public static void main(String[] args)

{

// 6. Statements

System.out.println("Hello, World!");

}

}

Explanation of Each Part

1. Comments

// This is a single-line comment

/*

This is a
 multi-line comment

*/

Purpose

- Used to explain the code.
- Ignored by the compiler.
- Makes the program easier to understand.

2. Package Declaration (Optional)

package mypackage;

Purpose

- Groups related classes together.
- Helps organize large projects.

Example

package studentmanagement;

3. Import Statements (Optional)

import java.util.*;

Purpose

- Allows use of predefined classes from Java libraries.

Example

```
import java.util.Scanner;
```

Here Scanner is used to take input from the user.

4. Class Declaration

```
public class HelloWorld
```

Purpose

- Every Java program must contain at least one class.
- The class name and file name should be the same.

Example

```
public class Student
```

File name:

Student.java

5. Main Method

```
public static void main(String[] args)
```

Purpose

- Entry point of a Java program.
- Execution starts from here.

Breakdown

Keyword	Meaning
---------	---------

public	Accessible to JVM
--------	-------------------

static	Can be called without creating an object
--------	--

void	Returns no value
------	------------------

main	Starting method
------	-----------------

String[] args	Command-line arguments
---------------	------------------------

6. Statements

```
System.out.println("Hello, World!");
```

Purpose

- Performs actions such as calculations, input, and output.

Breakdown

Part	Meaning
------	---------

System	Predefined Java class
--------	-----------------------

out	Standard output stream
-----	------------------------

println()	Prints output and moves to next line
-----------	--------------------------------------

"Hello, World!"	Text to be displayed
-----------------	----------------------

Output

Hello, World!

Complete Example Program

```
import java.util.*;
```

```
public class StudentInfo
```

```
{
```

```
    public static void main(String[] args)
```

```
    {
```

```
        String name = "Rahul";
```

```
        int age = 20;
```

```
        System.out.println("Student Name : " + name);
```

```
        System.out.println("Student Age : " + age);
```

```
}  
}
```

Output

Student Name : Rahul

Student Age : 20

Execution Flow of a Java Program

1. Write Program (HelloWorld.java)
↓
 2. Compile Program (javac HelloWorld.java)
↓
 3. Bytecode Generated (.class file)
↓
 4. JVM Executes Bytecode
↓
 5. Output Displayed
-

Quick Memory Trick

C → P → I → C → M → S

- **C** = Comments
- **P** = Package
- **I** = Import
- **C** = Class
- **M** = Main Method
- **S** = Statements

This is the standard structure followed by almost every Java program.